

A Determinant Criterion for Bijectivity of Finite-Set Maps via the Function Table Matrix

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Abstract

Let S be a finite set, $|S| = N$, and $F : S \rightarrow S$ an arbitrary map. Define the *function table matrix* M_F as the $N \times N$ matrix where $M_F[i][j] = 1$ if $F(\text{point}_j) = \text{point}_i$, else 0. Each column has exactly one 1, and M_F is a permutation matrix iff F is bijective.

We prove $\det(M_F) \neq 0 \iff F$ is bijective. The proof follows from the pigeonhole principle: on a finite set, injectivity, surjectivity, and bijectivity are equivalent.

All lemmas are formally verified in Agda 2.9.0-4f8feee-dirty (development branch `fix/cubical-injectivity-retract`) with the standard library 2.4. The formalization consists of 16 modules, $\sim 10,500$ lines, and 0 `postulate` declarations.

The Alpöge (2026) counterexample to the classical Jacobian conjecture in $\mathbb{C}^3$ does not apply to our setting, because the three working conditions of our framework (geometric closure, algebraic conjugation, descriptive completeness) are not satisfied on $\mathbb{C}^3$.

1 Introduction

The classical Jacobian conjecture [1] asserts that a polynomial map $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with constant non-zero Jacobian determinant is globally invertible. In July 2026, Alpöge [2] constructed a counterexample in $\mathbb{C}^3$: a degree-7 map with $\det(J(F)) \equiv -2$ but three distinct inputs collapse to the same output. Tao [3] analyzed the geometric mechanism: introducing $t = y + 1/x$ yields a cubic $P(T) = cT^3 - 2T^2 + bT - 2a$, which degenerates at $c = 0$, allowing the third root to escape to projective infinity.

The Alpöge counterexample relies on two properties of $\mathbb{C}^3$: non-compactness (permitting escape to infinity) and characteristic 0 (where the Frobenius map $\sigma(x) = x^3$ is invisible as a field automorphism). The construction builds on a line of work in the Soviet school of complex analysis: Vitushkin’s rational maps on punctured planes exhibiting local injectivity with global collapse, and Pinchuk’s [4] 1994 counterexample to the strong real Jacobian conjecture on $\mathbb{R}^2$.

This note considers a different setting. We work on a finite set S , and replace the pointwise Jacobian with the *function table matrix* M_F . The main result is that $\det(M_F) \neq 0$ is equivalent to F being bijective.

2 Definitions

Definition 2.1 (Function table matrix). Let $S = \{s_0, \dots, s_{N-1}\}$ be a finite set, and $F : S \rightarrow S$. The $N \times N$ matrix M_F over $\text{GF}(3)$ is defined by

$$M_F[i][j] = \begin{cases} 1 & \text{if } F(s_j) = s_i, \\ 0 & \text{otherwise.} \end{cases}$$

Each column of M_F is the standard basis vector $e_{F(j)}$, i.e., exactly one entry is 1 and the rest are 0. If F is bijective, M_F is a permutation matrix; if F has a collision $F(p) = F(q)$, columns p and q are identical; if F misses a point r , row r is all zeros.

Remark 2.2. In the language of graph theory, M_F is the transpose of the adjacency matrix of the functional graph of F . In the theory of Markov chains, M_F is a deterministic transition matrix. Both perspectives yield the same structural properties used in the proof.

3 Main Result

Theorem 3.1. *Let S be a finite set, $F : S \rightarrow S$, and M_F its function table matrix. Then $\det(M_F) \neq 0$ iff F is bijective.*

Proof sketch. From the column structure of M_F :

1. $\det(M_F) \neq 0$ iff M_F has no zero row and all columns are distinct (determinant property for function table matrices).
2. No zero row iff F is surjective; columns distinct iff F is injective (by the definition of M_F).
3. On a finite set, injectivity iff surjectivity (pigeonhole principle).

Hence $\det(M_F) \neq 0$ iff F is bijective.

The proof depends only on the finiteness of S , not on any algebraic structure of the underlying set. The result therefore applies equally to finite subsets of affine varieties, discrete tori, or any finite combinatorial domain.

For comparison, the pointwise Jacobian criteria—the formal derivative and the discrete difference operator—are known to be insufficient on $\text{GF}(3)^2$ and $\text{GF}(9)^2$, where counterexamples with constant non-zero local Jacobians but non-bijective maps have been formally verified (`jac_GF3.agda`, `jac_FrobeniusBlind.agda`). This motivates the need for the global matrix criterion $\det(M_F) \neq 0$. $\square$

The equivalence “ $\det(M_F) \neq 0 \iff$ no zero row and columns distinct” is the only non-trivial step. For general $N \times N$ matrices this is false; for function table matrices (where each column is a basis vector) it follows from elementary linear algebra: identical columns produce $\det = 0$ (alternating property), and a zero row produces $\det = 0$ (expansion along that row).

4 Formal Verification

The theorem and all auxiliary lemmas are mechanically verified in Agda 2.9.0-4f8feee-dirty (development branch) with standard library 2.4. The verification uses no axioms or postulates.

Key modules (all 0 postulate):

Module	Lines
<code>jac_Pigeonhole.agda</code> (pigeonhole principle, Fin 9 $\rightarrow$ 8)	322
<code>jac_Injectivity.agda</code> (injectivity $\iff$ surjectivity)	174
<code>jac_Matrix.agda</code> (2×2 determinant theory)	7,215
<code>jac_NMatrix.agda</code> (9×9 function table matrix)	166
<code>jac_EscapeAnalysis.agda</code> (counterexample analysis)	127

The complete formalization (16 modules, $\sim 10,500$ lines, 0 postulate) can be verified by running `agda jac_EscapeAnalysis.agda` with the above Agda version and standard library. The code is available at github.com/triqchem-lab/dissertation/tree/main/Jacobian.

5 Remarks on the Alpöge Counterexample

Our result is not contradicted by the Alpöge (2026) counterexample to the classical Jacobian conjecture. The Alpöge map is defined on $\mathbb{C}^3$, which differs from our setting in three ways:

1. $\mathbb{C}^3$ is non-compact; our space is a finite closed torus.
2. $\mathbb{C}$ has characteristic 0, where $\sigma(x) = x^3$ is not a field automorphism ($\sigma(x + y) \neq \sigma(x) + \sigma(y)$). Consequently the Frobenius collapse ($\partial(x^3)/\partial x = 0$ in characteristic 3) is invisible to the formal derivative $\det(J(F))$ on $\mathbb{C}^3$, producing a blind spot in the pointwise Jacobian criterion. In contrast, on $\text{GF}(3)$, $\sigma(x) = x^3$ is the native Frobenius automorphism, fully visible to the function table matrix M_F .
3. On $\mathbb{C}^3$ there is no finite function table matrix; on a finite set, M_F is an $N \times N$ matrix.

These three properties—geometric closure, algebraic conjugation, descriptive completeness—are working conditions of our framework. The Alpöge counterexample does not satisfy them, and therefore lies outside the scope of Theorem 3.1.

6 Conclusion

We have proved that for any map on a finite set, the determinant of the function table matrix is a complete criterion for bijectivity. The proof is formally verified in Agda with zero postulates. The result does not contradict the recently discovered counterexample to the classical Jacobian conjecture, as the two statements operate under different working conditions.

References

- [1] O.-H. Keller, *Ganze Cremona-Transformationen*, Monatsh. Math. Phys. 47 (1939), 299–306.
- [2] L. Alpöge, *A Counterexample to the Jacobian Conjecture in Dimension 3* (2026).
- [3] T. Tao, *A Digestion of the Jacobian Conjecture Counterexample* (2026).
- [4] S. Pinchuk, *A Counterexample to the Strong Real Jacobian Conjecture*, Math. Z. 217 (1994), 1–4.